

Operational Readiness for LLM Hallucinations: A Counting-Process Capacity Model and Measured Evidence on Guardrail Cascades

Working draft. Code, data pointers and analysis scripts: [hallucination_poisson/](#) in the accompanying repository.

Abstract

Organisations that deploy large language models (LLMs) must decide how much human triage capacity to hold in reserve and how many automated guardrail layers to place in front of the model. Two simple heuristics, provisioning at the mean incident load and crediting each guardrail layer with its stand-alone effectiveness as if layers failed independently, are easy to apply but can be improved on. We describe the capacity requirement as a quantile of the daily count of human-handled incidents and replace a single Poisson description of all failures by typed classes: hallucinations as thinned, over-dispersed streams; provider incidents as seasonal, level-adjusted processes with at most weak self-excitation; serving failures as heavy-tailed episodes; and heavy-tailed human service. We show analytically that over-dispersion inflates required capacity, that positive association between layers makes the independence formula a lower bound on residual risk, and that, because capacity is integer-valued, cost is not unimodal in the number of layers. We evaluate the pipeline on the outputs of three small open LLMs that answered the same 300 SQuAD 2.0 questions (900 answers) through three executed guardrail layers with measured latency and failures; user arrivals are simulated and informed by public traces of an LLM service, of LLM providers' incidents and of company ticket systems, none of which is a hallucination log (one curated database of dated court cases, legal domain, is the only partial exception). A single task-specific verifier, adopted after an NLI classifier proved uninformative and evaluated on the same questions, reduced required capacity by 78-95% but blocked 15-71% of correct answers; a routed judge, under a 2 s fail-open timeout, did not demonstrably pay for itself (paired cost-saving intervals \$429 [-\$5, \$931]; -\$2 [-\$5, \$0]; \$61 [-\$9, \$646] per day) and added about three times the latency. The share of hallucinations passing a two-layer cascade was 1.3-1.6 times the independence prediction without routing (intervals include 1 for two of three generators) and 1.9-2.3 times with routing. With mild burstiness (daily-rate CV 0.3-0.44) a Poisson-sized capacity plan was exceeded on 23-32% of days without guardrails and 7-16% with the best cascade, against a target of 5%; in public ticket streams the net-of-trend daily-rate CV was 0.10-0.38 after excluding holiday and low-volume days (0.25-0.41 with all days) and a causal Poisson plan was exceeded on 9% of work days for the two long streams, whereas over-dispersion changed the 95% capacity by at most one unit for low-rate streams (provider incidents, dated court cases). In an illustrative organisation, a typed superposition required 14-48% more daily capacity than a single Poisson on the total, with the gap driven by the assumed dispersion of the hallucination class and by how strongly over-dispersed the serving class is taken to be. Hallucination arrivals are simulated and, to our knowledge, no operator log of hallucinations is public; the models are small and the task is extractive question answering, so the numbers do not transfer to production, whereas the method and its diagnostics do.

Keywords: LLM operations, hallucination, guardrails, capacity planning, Poisson process, over-dispersion, design science

1. Introduction

Large language models (LLMs) are now embedded in customer-facing and decision-support information systems, where an ungrounded answer is a service incident rather than a quality blemish. IS research has examined adoption and value of generative AI, but the *operations* question that follows deployment has received little formal treatment: how much human triage and verification capacity must an organisation keep on standby, and how many automated guardrail layers should sit in front of the model, when hallucinations arrive at random?

Two simple planning heuristics are easy to apply but can be improved on: provisioning triage capacity at the mean incident load, and crediting each additional guardrail layer with its stand-alone effectiveness as if layers failed independently. Provisioning at the mean leaves the system under-provisioned on roughly 30-50% of days in our measurements, and the independence product is, as we show, an optimistic bound.

We make four contributions and are explicit about which parts apply known results. **First**, we describe the capacity requirement for LLM incident handling as the quantile of the daily count of human-handled events (Section 3). The quantile rule and the over-dispersion correction are classical (Whitt, 2007; Cameron and Trivedi, 1998); our contribution is to apply them to this arrival stream, to derive how over-dispersion and thinning by guardrails enter the requirement (Propositions 2 and 3), and to check on public request, incident and ticket traces whether the counting-process assumptions hold on comparable real streams. Apart from one curated database of dated court cases (legal domain), none of these traces is a hallucination log, so hallucination arrivals themselves are assumed or simulated. **Second**, we replace the single-Poisson description of "all failures" by a typed model (Section 3.6): hallucinations as thinned, over-dispersed streams; provider incidents as a seasonal, level-adjusted and at most weakly self-exciting process; serving failures as heavy-tailed episodes; human triage with heavy-tailed service; with superposition and clustering results (Proposition 6) and public-data fits (Section 5.8). **Third**, we restate the multiplicative guardrail formula as the independence special case of Fréchet-type bounds on an arrival-rate cascade (Proposition 4). Dependence between guardrail layers has been measured on real defence stacks (Alotaibi et al., 2026) and the independence product is known to over-credit redundancy (Bhardwaj et al., 2026); we add a measurement on hallucination-grounding layers with uncertainty intervals and carry the correction through to the capacity decision. We also show that, once capacity is integer-valued, total cost is generally *not* unimodal in the number of layers, with comparative statics in the labour-to-compute ratio (Proposition 5); this follows from the model structure and is a model property, not an empirical discovery. **Fourth**, we evaluate the pipeline on real model outputs: three small open models answering the same 300 SQuAD 2.0 questions, three guardrail layers executed with measured latency and error behaviour, and traffic replayed through a non-homogeneous arrival process whose daily-rate variability is informed by public traces. Per-query hallucination outcomes, guardrail decisions, layer dependence and latency are measured; arrival times are simulated. To our knowledge, no earlier study combines measured guardrail performance, an incident-count model with dispersion and clustering, and a quantile-based capacity decision in one framework evaluated on real system outputs and public production traces; each ingredient has precedents (Section 2).

The paper follows the design-science tradition in IS (Hevner et al., 2004; Gregor and Hevner, 2013): we propose an artifact (a capacity-planning and guardrail-sizing method), ground it in a formal model, and

evaluate it on real system outputs. We are explicit about what the evaluation does and does not establish (Section 6).

2. Related Work

Hallucination and its mitigation. Surveys document hallucination as a pervasive property of neural text generation (Ji et al., 2023). Mitigation work includes retrieval grounding (Lewis et al., 2020), sampling-based consistency checks (Manakul et al., 2023), safeguard classifiers (Inan et al., 2023) and LLM-as-judge verification (Zheng et al., 2023). This literature evaluates *detectors* by accuracy; it does not ask how detector performance translates into the human capacity an organisation must hold in reserve.

Capacity planning under random demand. The use of the Poisson quantile to size service capacity is classical in service operations (Whitt, 2007), and time-varying demand is handled by piecewise-stationary staffing (Green et al., 2007). Empirical studies of call centres report that arrivals are well described by a Poisson process only conditional on a random, day-specific rate, i.e. they are over-dispersed (Brown et al., 2005); count-data econometrics offers the negative-binomial remedy (Cameron and Trivedi, 1998). We import these tools to a new arrival stream and test whether the same caveats apply.

Human review capacity for AI outputs. Queueing models of human review of AI output exist: Poisson escalation arrivals with Erlang-C staffing of human operators for agent escalations (Besanson, 2026), M/G/1 models of reviewing AI drafts (Bartolucci and Vivo, 2026), and fluid-limit routing between an LLM judge and human review (Lin and Zhang, 2026). These use mean-based stability or waiting-time criteria and assume, rather than test, Poisson arrivals. We size capacity at a quantile of the incident-count distribution and test the arrival assumptions.

Counting-process models of AI failures. Public-health-style monitoring of AI incident counts has found strong over-dispersion relative to Poisson in incident-database counts of vehicle and deepfake incidents (Abraham et al., 2026), and Markov reliability models of LLM agents reduce to the Goel-Okumoto NHPP software-reliability model in the rare-failure regime (Tran-Truong and Le, 2026). Neither concerns hallucination or guardrail-bypass events, nor applies time-rescaling tests or capacity sizing.

Layered defences. The multiplicative formula for the residual rate of independent safeguards is standard in reliability engineering, and it over-credits redundancy when failures are positively associated (Esary et al., 1967). For LLM defence stacks, Alotaibi et al. (2026) measure positive failure correlation between seven defence layers (ϕ between 0.30 and 0.75 on JailbreakBench) and call the product a lower bound that no real stack attains; Bhardwaj et al. (2026) give a Fréchet-Hoeffding sandwich and measure high co-failure among same-model LLM components. Our dependence results should be read as additional evidence, in a different setting (hallucination-grounding layers), rather than as a first demonstration.

Gap. Prior work addresses these pieces separately: arrival models for AI incidents, human-review queueing, measured layer dependence, and guardrail cost-latency trade-offs. We integrate them for LLM hallucination operations and evaluate the integrated pipeline on measured outputs and public traces. The literature on LLM guardrails moved quickly in 2026; our search was not exhaustive and similar work may exist.

3. Model

3.1 Arrivals of critical hallucinations

Users send queries at times governed by an intensity $\Lambda(t)$ with a daily cycle. Each query independently produces a *hallucination* (a wrong or ungrounded answer) with probability p_0 , and the guardrail cascade blocks some of them. We assume that every hallucination that passes all guardrails is detected and becomes exactly one human-handled event that consumes one unit of triage capacity, and call such an event a *critical hallucination*; let $q \leq p_0$ be the probability that a query yields one. By the thinning theorem (Kingman, 1993), critical hallucinations form a non-homogeneous Poisson process (NHPP) with intensity $q\Lambda(t)$. Let X_d be the number of events on day d , with mean $\lambda = q \int_{\text{day}} \Lambda(t) dt$. Then $X_d \sim \text{Poisson}(\lambda)$ regardless of the within-day shape of Λ ; the within-day shape matters for *hourly* staffing, where the hour- h requirement is the quantile of $\text{Poisson}(\lambda_h)$. The detection assumption is strong: if only a fraction d of the surviving hallucinations is ever noticed and triaged, the event rate is $d\lambda$ and, by Proposition 3, the dispersion index is $1 + d\lambda c^2$; the capacities we report are therefore upper bounds that hold for $d = 1$, and detection itself is a further failure point that the model does not cover.

Inter-arrival times are exponential only under a constant rate. Under the NHPP, the time-rescaled gaps $\Lambda(t_i) - \Lambda(t_{i-1})$ are i.i.d. $\text{Exp}(1)$ (the time-rescaling theorem; Daley and Vere-Jones, 2003; Ogata, 1988), which gives a goodness-of-fit test that we use in Section 5. The memoryless property therefore applies to rescaled time: elapsed time since the last event carries no information, but time-of-day does.

If the daily rate is itself random, $\Lambda_d = \lambda M$ with $E[M] = 1$ and $\text{Var}(M) = c^2$, the daily count is a mixed Poisson variable with $\text{Var}(X_d) = \lambda + \lambda^2 c^2 = \lambda \phi$, where $\phi = 1 + \lambda c^2$ is the dispersion index. This is the Cox-process (doubly stochastic) alternative to the pure Poisson null (Cox, 1955; Cox and Isham, 1980).

3.2 Readiness capacity

For reliability target α , the capacity requirement is the generalised inverse of the daily-count distribution:

$$k^*(\alpha, \lambda) = \min \{ k \in \mathbb{N}_0 : P(X_d \leq k) \geq \alpha \}$$

Proposition 1 (Poisson capacity). Let $X \sim \text{Poisson}(\lambda)$. (a) $k^*(\alpha, \lambda)$ is non-decreasing in λ . (b) $k^*(\alpha, \lambda) = 0$ if and only if $\lambda \leq -\ln \alpha$; for $\alpha = 0.95$ this threshold is 0.0513 events/day. (c) Provisioning at the mean, $k = \lfloor \lambda \rfloor$, leaves the system under-provisioned with probability $P(X > \lfloor \lambda \rfloor)$, which is at least 0.26 for every $\lambda \geq 1$, is about 0.35 at $\lambda = 3$ and tends to 0.5 as λ grows. (d) $k^* = \lambda + z_\alpha \sqrt{\lambda} + O(1)$ as $\lambda \rightarrow \infty$.

Proof sketch. (a) $\text{Poisson}(\lambda)$ is stochastically increasing in λ . (b) $P(X = 0) = e^{-\lambda} \geq \alpha$ iff $\lambda \leq -\ln \alpha$. (c) On $[n, n+1)$ the probability $P(X > n)$ is increasing in λ , so its minimum over $\lambda \geq 1$ is $P(X > 1) = 0.264$ at $\lambda = 1$, and it is 0.353 at $\lambda = 3$ ($P(X \leq 3) = 0.647$). (d) follows from the Cornish-Fisher expansion of the Poisson quantile. ■

Proposition 2 (over-dispersion inflates capacity). Let $X_d | M \sim \text{Poisson}(\lambda M)$ with $E[M] = 1$ and $\text{Var}(M) = c^2 > 0$. Then (a) $\text{Var}(X_d) = \lambda \phi$ with $\phi = 1 + \lambda c^2$; (b) X_d dominates $\text{Poisson}(\lambda)$ in the convex order, so $E[(X_d - k)^+] \geq E[(Y - k)^+]$ for every k , where $Y \sim \text{Poisson}(\lambda)$; (c) If ϕ stays bounded as λ grows, $k^*_{\text{mixed}}(\alpha) = \lambda + z_\alpha \sqrt{\lambda \phi} + O(1)$. If instead the multiplier has a fixed CV c , then $\phi = 1 + \lambda c^2$ grows with λ and $k^*_{\text{mixed}}(\alpha) = \lambda \cdot F_M^{-1}(\alpha) + O(\sqrt{\lambda})$, where F_M is the distribution of M ; in that regime $\lambda + z_\alpha \sqrt{\lambda \phi}$ understates the exact quantile by an amount that grows linearly in λ , and $\sqrt{\phi}$ is a lower guide for the inflation of the safety stock $k^* - \lambda$, not an exact

factor (e.g. for Gamma M at $\lambda = 120.7$ and $c = 0.3$ the exact ratio of the safety stock to its Poisson value is 3.7 against $\sqrt{\phi} = 3.4$; at $c = 0.44$ it is 5.5 against 4.9). The exact negative-binomial quantile (Table 1) should be used for planning.

Proof sketch. (a) is the law of total variance. (b) For Poisson, $\partial/\partial\lambda E[(Y - k)^+] = P(Y \geq k)$, which is non-decreasing in λ , hence $\lambda \mapsto E[(Y_\lambda - k)^+]$ is convex; Jensen's inequality applied to the mixing variable M gives $E[(X_d - k)^+] = E[g(\lambda M)] \geq g(\lambda)$. (c) For bounded ϕ the count is asymptotically normal with variance $\lambda\phi$. For fixed c , $X_d/\lambda \rightarrow M$ in distribution as $\lambda \rightarrow \infty$, so the 95% quantile of X_d is asymptotically $\lambda F_M^{-1}(\alpha)$ and the normal approximation fails. ■

Table 1 evaluates Proposition 2 exactly (negative-binomial mixing). At $\lambda = 10$ events/day and $c = 0.3$, the Poisson plan $k^* = 15$ is exceeded on 11.0% of days instead of 4.9%, and the correct requirement is 18; at $\lambda = 25$ and $c = 0.3$ the Poisson plan of 33 is exceeded on 16.7% of days and 41 is required.

Table 1. Capacity under over-dispersed arrivals (Gamma-mixed Poisson, $\alpha = 0.95$)

λ (events/day)	c (CV of daily rate)	ϕ	k^* Poisson	k^* mixed	P(exceed) Poisson plan
10	0	1.00	15	15	4.9%
10	0.1	1.10	15	16	5.7%
10	0.2	1.40	15	17	8.0%
10	0.3	1.90	15	18	11.0%
10	0.5	3.50	15	21	16.4%
25	0	1.00	33	33	5.0%
25	0.1	1.25	33	35	7.0%
25	0.2	2.00	33	37	11.9%
25	0.3	3.25	33	41	16.7%
25	0.5	7.25	33	50	23.0%

Exact negative-binomial quantiles; computed by `paper_tables.py`.

3.3 Thinning: guardrails make the incident stream closer to Poisson

Requests may be clustered even when incidents are rare. Because critical hallucinations are an independent thinning of requests, the dispersion of the incident stream follows from that of the request stream:

Proposition 3 (thinning reduces dispersion). Let N be the request count in a window with mean μ and Fano factor $F = \text{Var}(N)/\mu$, and let each request independently become a critical event with a constant probability p that does not depend on the request stream. Then the event count N_p has mean $p\mu$ and Fano factor $1 + p(F - 1)$. In particular, a guardrail cascade that reduces the residual probability p pulls the incident stream towards Poisson (Fano factor 1), while the unprotected stream (p large) retains most of the request stream's burstiness. Equivalently, for a mixed-Poisson day with rate multiplier of CV c , $\phi = 1 + \lambda c^2$ in Proposition 2 with $\lambda = p\mu$. Under independent thinning with constant p the multiplier M , and hence its CV c , is unchanged, so c (rather than ϕ) is the quantity that can be carried over between streams with different event rates; this transfer requires that p is constant and independent of M , which we assume but do not test for hallucinations.

Proof. $\text{Var}(N_p) = p^2\text{Var}(N) + p(1 - p)\mu$ by the law of total variance for the binomial thinning. Dividing by $p\mu$ gives $pF + (1 - p) = 1 + p(F - 1)$. ■ (We also verified this numerically by simulation.) The practical reading: over-dispersion matters most precisely where capacity is largest, i.e. without or with weak

guardrails, and matters less after strong filtering, consistent with the shrinking gap between Poisson and negative-binomial capacity in Section 5.4.

3.4 Guardrail cascades and dependence

A cascade of N layers with stand-alone clearance rates c_i (the share of critical hallucinations layer i blocks) is usually assumed to give $\lambda_N = \lambda_0 \prod (1 - c_i)$. This presumes independent layer decisions. In practice the same ambiguous or adversarial outputs tend to fool all layers.

Proposition 4 (dependence bounds). For any joint behaviour of the layers, $\lambda_0(1 - \sum c_i)^+ \leq \lambda_N \leq \lambda_0(1 - \max_i c_i)$. If the layers' catch indicators, conditional on a hallucination, are positively associated (Esary et al., 1967), then additionally $\lambda_N \geq \lambda_0 \prod (1 - c_i)$. Consequently the multiplicative formula under-states residual risk and, through Proposition 1(a), under-states required capacity.

Proof sketch. The bounds are the Fréchet-Bonferroni bounds for the event that no layer fires. For associated indicators, $P(\cap \{I_i = 0\}) \geq \prod P(I_i = 0)$ (Esary et al., 1967). ■

Table 2 illustrates the width of the identification problem with $c_i = 0.63$ and $\lambda_0 = 10$: without knowing the dependence, three layers could imply anything from 0 to 7 units of capacity, which is why the dependence must be *measured* (Section 5.2).

Table 2. Residual rate and capacity bounds ($\lambda_0 = 10$, $c_i = 0.63$, $\alpha = 0.95$)

N	λ_N lower	λ_N independent	λ_N upper	k* lower	k* indep.	k* upper
1	3.70	3.70	3.70	7	7	7
2	0.00	1.37	3.70	0	4	7
3	0.00	0.51	3.70	0	2	7
4	0.00	0.19	3.70	0	1	7

3.5 Cost-minimising guardrail depth

Let C_h be the daily cost of one unit of triage capacity, g_i the daily compute cost of layer i , $k_{\min} \geq 0$ a minimum standing capacity (an on-call engineer is not removed because incidents are rare), and consider the cost

$$f(N) = C_h \cdot \max\{k_{\min}, k^*(\alpha, \lambda_N)\} + \sum_{i \leq N} g_i \quad (+ \text{latency-feasibility: } \sum L_i \leq SLA).$$

Proposition 5 (optimal depth). (a) f is in general not unimodal in N : because k^* is integer-valued with plateaus, f can rise and then fall again. (b) With $k_{\min} \geq 1$, let N_0 be the first N with $k^*(\alpha, \lambda_N) \leq k_{\min}$; every $N > N_0$ is strictly dominated, so $N \leq N_0$. (c) Let $\theta = C_h / g$ with equal per-layer compute cost g . The minimiser $N(\theta)$ is non-decreasing in θ : when human triage is relatively expensive, deeper cascades are optimal (Topkis, 1978). (d) The latency constraint restricts N to a feasible prefix of the cascade.

Proof sketch. (a) is shown by the example in Table 3. (b) For $N > N_0$, labour cost is constant at $C_h k_{\min}$ while compute increases. (c) Write $f(N) = g[N + \theta K(N)]$ with K non-increasing; the difference $f(N+1) - f(N) = g[1 + \theta(K(N+1) - K(N))]$ is non-increasing in θ , so the largest minimiser is non-decreasing in θ by Topkis' monotone comparative statics. (d) is immediate. ■

The number of layers is an outcome of the model, not a constant. Table 3 uses an illustrative parameter set ($\lambda_0 = 10$ events/day, a clearance of 0.63 per layer under independence, \$150 per capacity unit, \$25 per layer). With $k_{\min} = 0$ the cost falls from $N = 3$ to $N = 4$, rises at $N = 5$ and falls again at $N = 6$, where k^* reaches zero ($\lambda_6 = 0.026 < -\ln 0.95$), so the minimum is at $N = 6$; with a realistic floor of one on-call engineer ($k_{\min} = 1$) the minimum is at $N = 4$. A fixed number of layers (for instance three) is therefore not implied by the cost model: the optimum depends on the cost ratio, the clearance rates, the dependence between layers and the latency constraint.

Table 3. Daily cost by guardrail depth ($\lambda_0 = 10$, $c = 0.63$ independent, $C_h = \$150/\text{unit}$, $g = \$25/\text{layer}$)

N	λ_N	k^*	Labour	Compute	Total ($k_{\min} = 0$)	Total ($k_{\min} = 1$)
0	10.000	15	\$2,250	\$0	\$2,250	\$2,250
1	3.700	7	\$1,050	\$25	\$1,075	\$1,075
2	1.369	4	\$600	\$50	\$650	\$650
3	0.507	2	\$300	\$75	\$375	\$375
4	0.187	1	\$150	\$100	\$250	\$250
5	0.069	1	\$150	\$125	\$275	\$275
6	0.026	0	\$0	\$150	\$150	\$300
7	0.009	0	\$0	\$175	\$175	\$325

Minimum: $N = 6$ under $k_{\min} = 0$; $N = 4$ under $k_{\min} = 1$. The cost parameters are illustrative assumptions.

3.6 Beyond a single Poisson: typed failure classes

LLM operations produce several kinds of failures that violate the Poisson assumptions in different ways, so treating "all failures" as one Poisson stream is a modelling error rather than a simplification. We therefore model each class with the process that matches its mechanism and combine the classes only at the level of the human-handled event (an incident or ticket). Table 4 summarises the mapping, the assumption that fails and the evidence developed in Section 5.8.

Table 4. Failure classes, mechanisms and models

Class	Mechanism	Poisson assumption violated	Model used	Evidence
Hallucination (per answered query)	Independent errors of the model given the query, on a clustered, daily-varying request stream	Constant rate; independence of requests	Cox / negative-binomial thinning of the request stream (Props. 2, 3)	Sections 5.4, 5.6, 5.7
Provider-declared incidents	Outages, possibly followed by further incidents within hours; weekday/hour seasonality and a drifting level	Orderliness and independence (self-excitation)	Seasonal (level-adjusted) Hawkes process, exponential kernel; NHPP as null	Section 5.8
Serving errors and timeouts	Common-cause episodes (outages, rate limiting) with heavy-tailed size; residual isolated errors (not modelled)	Orderliness (batch arrivals), independence	Episode process (Hawkes) with heavy-tailed marks; compound Poisson for counts of failed requests	Section 5.8
Human triage service	Mixture of quick and very long tickets	Exponential service	Heavy-tailed (Weibull / lognormal) service; workload headroom	Section 5.7

Proposition 6 (superposition and clustering). (a) If the daily counts of independent failure classes j have means λ_j and dispersion indices $\phi_j = \text{Var}/\text{mean}$, the total daily count has mean $\Sigma \lambda_j$ and $\text{Var} = \Sigma \lambda_j \phi_j$, so the 95% capacity is approximately $\Sigma \lambda_j + z \cdot \sqrt{\Sigma \lambda_j \phi_j}$. A single Poisson on the total uses $\phi = 1$ and under-states the safety stock by the factor $\sqrt{\phi_{\text{eff}}}$ with $\phi_{\text{eff}} = \Sigma \lambda_j \phi_j / \Sigma \lambda_j$. The superposition of independent classes is Poisson only if every class is Poisson. (b) For a stationary Hawkes process (Hawkes, 1971) with branching ratio $\alpha < 1$ (the expected number of direct follow-up events per event), the cluster started by a background event has a Borel size distribution with mean $1/(1 - \alpha)$, the long-window Fano factor of counts is $1/(1 - \alpha)^2$, and the count variance in a window of fixed length approaches this value as the window grows relative to the excitation time scale $1/\beta$. (c) For a compound Poisson count $N = \Sigma_{i \leq M} X_i$ with $M \sim \text{Poisson}(v)$, the Fano factor is $E[X^2]/E[X]$, which is unbounded when the mark distribution has an infinite second moment (e.g. a Pareto tail index below 2): the variance of the daily count, and with it the Poisson capacity error, is dominated by the largest episodes.

Proof sketch. (a) Variances of independent counts add; the normal approximation of the quantile uses the total variance. (b) The Hawkes cluster is a Galton-Watson tree with $\text{Poisson}(\alpha)$ offspring; its total progeny is $\text{Borel}(\alpha)$ with mean $1/(1 - \alpha)$ and second moment $1/(1 - \alpha)^3$, hence $E[X^2]/E[X] = 1/(1 - \alpha)^2$, and the stationary count variance-to-mean ratio of a Hawkes process equals this value in the long-window limit. (c) is Wald's identity for the compound Poisson variance, $\text{Var}(N) = vE[X^2]$. ■ For provider incidents the fitted branching ratio is 0.11-0.21 with a constant level and 0.05-0.09 once a monthly level is allowed (Section 5.8), implying a long-window Fano factor of about 1.3-1.6 and 1.1-1.2 respectively, to be compared with stratified dispersion indices of 1.2-1.5 (Section 5.6); for serving-failure episodes α is 0.5-0.85 depending on the episode definition and the second-moment ratio $E[X^2]/E[X]$ of the episode sizes is in the thousands, so the Fano factor of failed-request counts is dominated by a few large episodes and a Poisson description of failed requests is inadequate by several orders of magnitude, also after conditioning on traffic volume.

4. Empirical Design

4.1 Data, models and ground truth

We use the SQuAD 2.0 validation set (Rajpurkar et al., 2018), which combines answerable questions with questions that the supplied passage cannot answer. One random sample of 300 questions (seed 0; 148 unanswerable) is drawn once and given to all three models, so the three runs are paired on the same questions and are not independent samples. Each model receives the passage and the question and must reply with a short answer or the token UNANSWERABLE. A response is labelled a **hallucination** if it is not an abstention and either (i) the question is answerable and the answer matches no gold answer (normalised containment or token-F1 ≥ 0.6) or (ii) the question is unanswerable. Abstentions are never counted as hallucinations and bypass the guardrails. The generators are Qwen2.5-0.5B-Instruct, SmolLM2-360M-Instruct and Qwen2.5-1.5B-Instruct (Qwen Team, 2024; Allal et al., 2025), run greedily on CPU. We use small open models because they can be run end-to-end with full control and measured latency; their hallucination rate is higher than that of frontier models, so absolute rates should not be transferred (Section 6).

4.2 Guardrail layers

The three layers follow the asymmetric deterministic, classifier and semantic-judge architecture of the production literature, but each is implemented and executed on every answer:

- **L1, deterministic.** Regular expressions for e-mail, SSN, card and phone patterns; template-token leakage; degenerate repetition; and a grounding heuristic that blocks answers containing a number absent from the evidence passage.
- **L2, classifier (extractive-QA verifier).** A SQuAD 2.0-trained extractive-QA model (deepset/roberta-base-squad2) answers the question from the evidence passage; the score is the token-F1 between its span and the generator's answer (0 if it predicts "no answer"), and the output is blocked if the score is below 0.5 (a threshold we did not tune). The first implementation of L2 was an NLI cross-encoder (DeBERTa-v3-small; He et al., 2021) scoring $P(\text{passage entails "the answer to Q is A"})$; it was uninformative and was replaced after we had seen the results (Section 5.1). Llama Guard-type classifiers target safety-policy violations rather than factual grounding and were therefore not used.
- **L3, semantic judge.** An LLM judge (Qwen2.5-1.5B-Instruct in every run) given the passage, question and answer, replying SUPPORTED or UNSUPPORTED (Zheng et al., 2023). It is invoked only when routed: when L2 is present, only if the L2 score is below 0.9.

All layers use the same evidence passage as retrieval-grounded verification. Every layer is run on every example so that any subset or cascade can be evaluated offline from observed flags, without any independence assumption. We record flag, score, wall-clock latency and exceptions per call. Layers fail open: a call that raises an exception or takes longer than 2 s is counted as a system failure (from which we estimate $\lambda_{\text{sys_fail}}$ rather than assuming it), its verdict is discarded so that the output passes that layer, and the added latency is capped at 2 s. No exception occurred in any run, so all system failures are judge calls slower than 2 s (25-30% of the judge calls on our CPU setup). Section 5.5 gives a sensitivity analysis in which slow verdicts are waited for and credited.

4.3 Traffic replay and estimators

Production logs are unavailable, so *arrivals* are simulated while *outcomes* are measured. Queries arrive according to an NHPP with a 24-slot hour-of-day intensity and a mean of 200 queries/day for 90 days. The default profile is business-hours-shaped; a second profile is the hour-of-day profile measured in the BurstGPT production trace (Section 4.5). A Cox variant multiplies each day's intensity by a Gamma factor with mean 1 and coefficient of variation c to represent bursty demand; we use $c = 0.3$ and $c = 0.44$ as scenario values. Net of trend and after excluding holiday and low-volume days, public human-handled ticket streams give 0.10-0.38 (0.25-0.41 with all days; Section 5.7), so $c = 0.3$ lies within that range and $c = 0.44$ above it; 0.44 is the smallest coefficient of variation of any 14-day window of BurstGPT (Section 5.6), whose variability over the whole trace is much larger, so it is a floor for that trace and not a net-of-trend estimate comparable to the ticket values. Each arriving query replays a uniformly drawn example from the measured pool; it becomes a critical event (a hallucination that survives the cascade and is detected, Section 3.1) if it is a hallucination that survives the cascade. For each resulting event stream we estimate $\hat{\lambda}$ = events/days and test: (i) the index-of-dispersion test on daily counts; (ii) a chi-square test of constant hourly intensity; (iii) a Kolmogorov-Smirnov test of raw inter-arrival times against an exponential, and of time-rescaled gaps under the fitted hourly intensity (the Lilliefors-type p-value is optimistic); (iv) a negative-

binomial fit of daily counts. Uncertainty from the finite evaluation pool is quantified by a 300-draw bootstrap over examples (95% percentile intervals).

Because the same pool generates all arrivals, the replay tests whether the *counting process* has the claimed structure given measured per-query outcomes; it does not test whether real users' queries are independent. That is the main external-validity limit (Section 6).

4.4 Cascade selection and cost model

The replay evaluates the cascades L1, L2, L3 and their combinations in the canonical order L1, L2, L3 (all subsets, including no guardrail). **Selection objective.** The *best cascade* is the one that minimises $k^* + 0.02 \cdot (\text{mean added latency in ms}) + (\text{system failures per day})$ among cascades whose mean added latency is within the stated budget (100 ms, or 500 ms where indicated). The weights (1, 0.02 per ms, 1 per daily system failure) are arbitrary and are not tuned; they are used to produce one ranking for Sections 5.4 and 5.8 (L1+L2 under the 500 ms budget for all three generators). **Cost model.** Daily cost is $C_h \cdot \max(k_{\min}, k^*) + \text{compute} + \text{false-block cost} + \text{residual-harm cost}$, with $C_h = \$150$ per capacity unit, compute \$0.0001 per classifier call and \$0.002 per judge call, false-block cost \$1 per wrongly blocked answer and residual-harm cost \$100 per hallucination reaching a user (illustrative scenario values, not estimates). The *minimum-dollar cascade* of Section 5.5 minimises this cost; it can differ from the best cascade because it ignores latency and system failures. A sensitivity grid varies daily traffic (50, 200 and 800 queries), the false-block cost (\$0, \$1, \$10) and the residual-harm cost (\$0, \$100, \$1,000), i.e. 27 scenarios per generator. Uncertainty from the finite evaluation pool is quantified by pool-bootstrap intervals (resampling examples), including paired intervals for the cost differences between cascades.

4.5 Public production data

Public data from six sources test the counting-process assumptions on real systems: a request trace of an LLM service, the incident history of LLM providers, three company ticket logs, and a curated database of dated hallucination cases in court decisions. Only the last is a (legal-domain, curated) hallucination log; no operator log of hallucinations is public to our knowledge. **BurstGPT** (Wang et al., 2025; CC BY 4.0) contains 10.6 million request records of regional Azure OpenAI ChatGPT and GPT-4 services at one-second resolution over 231 days (two contiguous segments of 121 and 110 days separated by an unrecorded gap of 104 days; timestamps are seconds from an undisclosed day 0, so weekday labels are unknown); failed requests are the rows with zero response tokens. We estimate the hour-of-day profile, the day-to-day variability of request volume (robustly, because volume grows by two orders of magnitude and contains regime shifts), and the within-hour Fano factor of counts at 1 s, 10 s, 1 min and 10 min. **LLM Service Outages and Incident Reports** (Chu et al., 2025; CC BY 4.0) contains provider-declared incidents scraped from the OpenAI, Anthropic and Character.AI status pages; we use the 457 incidents starting from March 2023, when all three pages are populated, and test Poisson counts per day and week (index of dispersion, also stratified by day of week), day-of-week and hour-of-day homogeneity, exponential inter-arrival times after time-rescaling with a fitted hour-of-week intensity, the distribution of incident durations as a proxy for human response time, and the number of concurrently open incidents against the M/G/ ∞ prediction. For **human triage workloads** we use three public company ticket datasets: customer tickets of an international software company (Abdellatif, 2026; 22,016 customer tickets from 2017 to March 2023, with creation, resolution and in-progress times), the audit log of a ServiceNow instance of an IT company (Amaral et al., 2018; 24,918 incidents in 2016-2017) and the help-desk log of an Italian software company (Verenich,

2016; 3,804 cases, 2010-2012). Filters: for the software company we keep tickets of type "Ticket", created in 2017 or later and resolved as "Done" (22,016 of 66,691 rows; 45,275 are of type Ticket, 23,871 of them created from 2017 on); the work week is detected from the data as the five busiest days of the week (Sunday-Thursday for this company, Monday-Friday for the others). The ServiceNow log has 98.9% of its incidents opened between 29 February and 1 June 2016, so its arrival analysis uses only the middle weeks (7 March to 27 May 2016) and its concurrency is not analysed. Holidays and bridge days produce isolated low-count days that a local mean cannot absorb, so each arrival analysis is reported twice: for all work days, and with such days excluded (software company and ServiceNow: work days with fewer than 35% of the centred 41-work-day rolling median, a rule that is reliable only for high-rate streams, 62 and 5 days; Italian help desk: Italian national holidays and Easter Monday, 26 days; court cases: US federal holidays, 10 days). For each stream we estimate weekday-stratified over-dispersion, the daily rate CV net of slow trends (Pearson dispersion of weekday counts against a leave-one-out local mean over about eight weeks, after removing the weekday effect), the exceedance rate of Poisson-sized capacity plans, the distribution of resolution and in-progress times (exponential, lognormal and Weibull fits compared by AIC), the daily workload safety factor (95th percentile over mean of handling hours versus arrivals), and the number of concurrently open tickets against the $M/G/\infty$ Poisson prediction. For **dated hallucination incidents** we analyse the Charlotin database of court decisions with AI-fabricated content (Charlotin, 2026): the decision date at day resolution for October 2025 to August 2026, work-day counts with the same stratified and local-mean analysis as for tickets. The Poisson plan is evaluated twice: against a two-sided local mean over about 40 surrounding work days (in-sample, not a feasible forecast) and causally, with the trailing 20-work-day mean and a coefficient of variation estimated on earlier days only (rolling origin); the weekday effects and the work-week definition are estimated on the full sample, so only the level and the CV are strictly causal. Dated cases of the most recent months are excluded because of reporting lag (September 2026, 95 cases, and one case in October 2026).

Typed failure models. For the provider incidents we fit, by maximum likelihood, a seasonal Hawkes process with an exponential kernel, $\lambda(t) = \mu_0 s(t) + \sum \alpha \beta e^{-\beta(t - t_i)}$, where $s(t)$ is a weekday/weekend $\times$ hour-of-day profile estimated from the data and held fixed, and compare it with the seasonal NHPP ($\alpha = 0$) by AIC, by a parametric-bootstrap likelihood-ratio test (200 simulations under the NHPP; the boundary $\alpha = 0$ invalidates the usual χ^2 reference), and by the time-rescaling test using the fitted compensator. The estimator was checked by simulation (recovery of μ_0 , α , β from synthetic seasonal Hawkes data, and no false excitation when $\alpha = 0$). For BurstGPT failures we detect outage-like episodes (minutes with more than 20% failed requests and at least five failures, merged when at most ten minutes apart), characterise episode duration, size (lognormal fit, Hill tail index) and arrival process (daily dispersion, seasonal Hawkes fit), and test four alternative episode definitions (five including the main one). Finally, we combine class-specific negative-binomial daily counts (means and dispersion indices taken from the estimates above, classes independent; 400,000 simulated days, seed 0) by Monte Carlo to compare the 95% capacity of the typed superposition with a single Poisson on the total (Section 5.8); the hallucination class uses the measured rate of the L1+L2 cascade for Qwen2.5-0.5B and an assumed daily-rate CV in 0.3-0.45, the provider class the rate and stratified dispersion index of Section 5.6, and the serving class the episode rate over the 231 recorded days with the daily dispersion index under three treatments (two episode definitions and a volume-conditional value). All simulations and bootstrap draws are seeded and reproducible with the scripts of the repository.

5. Results

5.1 Hallucination rates and guardrail layer performance

On the same 300 SQuAD 2.0 questions given to each model (148 of them unanswerable), the hallucination rate is 60.3% (Qwen2.5-0.5B), 65.0% (SmolLM2-360M) and 47.0% (Qwen2.5-1.5B); 75-90% of the hallucinations are answers to unanswerable questions. Table 5 reports the layers. **Layer L1** (rules) catches 0.7-5.1% of hallucinations: this task contains no PII or template leaks and few fabricated numbers, so the deterministic layer is nearly inert here, and nearly free of latency and false blocks (one false block in 119 correct answers for Qwen2.5-0.5B, none for the others). **Layer L3** (the 1.5B judge) blocks 48-59% of hallucinations and 17-24% of correct answers, and each call takes about 1.8-1.9 s on our CPU. 25-30% of the judge calls exceed the 2 s timeout; under the fail-open policy of Section 4.2 their verdicts are discarded, which is why these clearances are lower than they would be if slow verdicts were credited (Section 5.5 gives that sensitivity). The judge is Qwen2.5-1.5B-Instruct in every run, so the Qwen2.5-1.5B arm is judged by itself; we did not test for a self-judging effect.

Negative result for the NLI classifier. Our first implementation of L2 was an NLI cross-encoder. It was essentially uninformative: its entailment score separates correct from hallucinated answers with AUC 0.52-0.58 (0.5 is chance), and at the default threshold it blocked 87-88% of hallucinations and 85-90% of *correct* answers. Its apparent capacity savings were therefore an artefact of blocking nearly everything: a capacity reduction by itself does not distinguish a useful layer from a layer that blocks indiscriminately, which is why false-block rates are reported next to it. We replaced the NLI layer by an extractive-QA verifier (a SQuAD 2.0-trained RoBERTa model whose own answer span is compared with the generator's answer by token-F1; AUC 0.84-0.90) and re-ran all layers on the stored answers. This replacement was made after seeing the results and the verifier is evaluated on the same 300 questions; it is effectively a second solver of the task it verifies and it was trained on SQuAD, which flatters it relative to a generic classifier. All numbers below use it.

Table 5. Guardrail layers on the same 300 SQuAD 2.0 questions for each generator (CPU inference, latency re-measured without concurrent jobs)

Generator	p_0 (hallucination rate)	c_1 (L1)	c_2 (L2 QA)	c_3 (L3 judge)	L2 false-block (good answers)	L3 false-block	L2 mean ms	L3 mean ms (capped at 2 s)
Qwen2.5-0.5B	60.3%	2.8%	89.5%	58.0%	15.1%	24.4%	98	1759
SmolLM2-360M	65.0%	5.1%	96.9%	48.2%	70.9%	16.5%	97	1792
Qwen2.5-1.5B	47.0%	0.7%	81.6%	58.9%	15.4%	22.8%	95	1728

c_i = stand-alone share of hallucinated answers the layer blocks. False-block = share of correct (non-hallucinated, non-abstained) answers blocked (denominators 119, 103 and 136). The L2 threshold ($F1 < 0.5$) was not tuned; SmolLM2 gives longer answers, which lowers its token-F1 agreement and inflates its L2 false-block rate.

5.2 Dependence between layers and the cost of routing

Among hallucinated answers the catch decisions of L2 and L3 are weakly positively associated ($\phi = 0.15, 0.11, 0.12$ for the three generators; 95% bootstrap intervals [0.00, 0.29], [-0.02, 0.21], [-0.06, 0.28]; Fisher exact $p = 0.054, 0.213, 0.186$), in the direction of Proposition 4. The intervals include zero for two of the three generators and the three runs use the same questions, so they are not independent replications; the

evidence is suggestive, not conclusive. L1 catches only 1-10 hallucinations per generator, so the L1 pairs ($\phi \leq 0.09$) are uninformative. Consequently, the excess of the residual share over the independence formula is modest (Figure 1, left). For the L2+L3 cascade, the share of hallucinations that pass all layers is 1.6-7.6% under the independence formula (computed from the same sample's marginals) and 2.6-9.9% when both layers are run on every answer, i.e. 1.5, 1.6, 1.3 times higher (12, 5, 14 hallucinations pass; 95% pool-bootstrap intervals [1.0, 2.0], [0.9, 2.1], [0.8, 1.7]); the interval includes 1 for 2 of the three generators. The cost of dynamic routing, in which only low-agreement answers are sent to the judge, is clearer: the ratios are 2.1, 1.9, 2.3 ([1.7, 2.6], [1.7, 2.2], [1.9, 3.0]) and the number of passing answers rises to 17, 6, 25. Routing reduces judge calls by 89-92%, but the judge no longer sees the answers that the verifier considers consistent, and those contain a substantial share of the hallucinations that the judge would catch. The absolute numbers of passing answers are small, so all these estimates are imprecise.

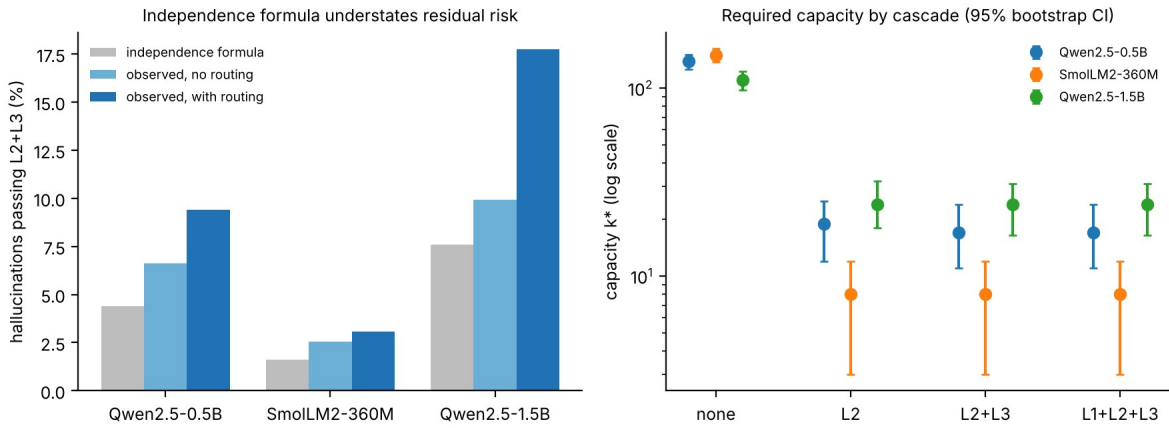


Figure 1. Left: share of hallucinations passing the L2+L3 cascade under independence, observed without routing, and observed with routing. Right: capacity k^* by cascade (95% bootstrap CI over the evaluated pool); log scale.

5.3 Counting-process checks under replayed traffic

With non-homogeneous Poisson arrivals and no burstiness, the replay reproduces the Poisson structure, as it must by construction: for the unprotected stream, the daily-count dispersion test does not reject ($p = 0.53$ - 0.91), the raw inter-arrival times are strongly non-exponential ($p < 10^{-100}$) because the intensity varies with the hour (hour-of-day homogeneity rejected), and the *time-rescaled* gaps are consistent with $\text{Exp}(1)$ ($p = 0.50$ - 0.85). We stress that this is a check that the estimators and the replay are correct and that the NHPP/rescaling machinery behaves as theory predicts; it is not evidence that real user traffic is a Poisson process, because the arrivals are simulated (Section 4.3). Within-day non-homogeneity matters for staffing by hour but not for the daily count.

5.4 Burstiness: Poisson-sized plans are exceeded under Cox traffic

When the daily intensity is multiplied by a Gamma factor with a coefficient of variation of only 0.3, the daily-count dispersion test rejects overwhelmingly ($p < 10^{-90}$ for the unprotected stream) and the negative-binomial fit gives r between 12.7 and 13.4. The Poisson capacity $k^* = 136, 146, 108$ (for the three generators, no guardrails) is then exceeded on 23-28% of days instead of the targeted 5%; the correct negative-binomial requirement is 178, 194, 141, i.e. 31-33% more (Figure 2; Proposition 2). Under the best cascade by the objective of Section 4.4 (L1+L2, $\lambda \approx 4$ -17/day) the effect is smaller in absolute capacity but the observed exceedance of the Poisson plan is still 10-11% (Table 6). This replay follows from the injected

Gamma multiplier and so confirms Proposition 2 and Table 1 on measured outcomes rather than providing an independent empirical finding; the value $c = 0.3$ is a scenario, not an estimate. Provisioning at the mean ($k = \text{round of the replay's } \lambda$) is exceeded on 28-49% of days across all configurations and replays of Sections 5.3-5.4, against the 5% target of the quantile plan. The replay k^* values use each run's own estimated λ (about 117-121 events/day without guardrails), whereas Table 7 uses the pool rate (120.7), so the same cascade can show k^* values that differ by a few units between tables and Figure 2.

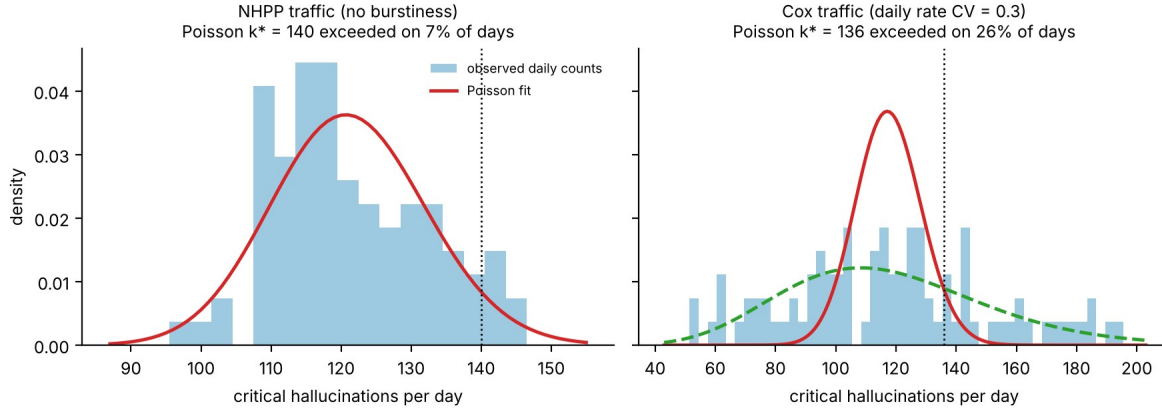


Figure 2. Daily critical hallucinations without guardrails (Qwen2.5-0.5B, 90 days, 200 queries/day). Left: steady NHPP traffic; right: bursty traffic. Dotted line: Poisson k^* ($\alpha = 0.95$).

Data-informed scenario. The smallest coefficient of variation of any 14-day window of the BurstGPT trace is $c = 0.44$ (a floor for that trace, whose median window CV is 0.91; Section 5.6), while the daily-rate CV net of trend in public human-handled ticket streams is 0.10-0.38 (Section 5.7; 0.25-0.41 before holiday and low-volume days are excluded). The scenario value $c = 0.3$ lies within the range of the ticket streams and $c = 0.44$ above it, at the BurstGPT floor. Replaying our measured outcomes with $c = 0.44$ and the BurstGPT hour-of-day profile gives the exceedance rates in Table 6.

Table 6. Share of days on which the Poisson-sized capacity k^* is exceeded (target 5%; 90 simulated days)

Generator	Cascade	Steady NHPP ($c = 0$)	$c = 0.3$	$c = 0.44 + \text{BurstGPT profile}$	k^* Poisson $\rightarrow k^*$ negative-binomial ($c = 0.44$)
Qwen2.5-0.5B	none	6.7%	25.6%	32.2%	135 $\rightarrow$ 209
Qwen2.5-0.5B	L1+L2	3.3%	10.0%	14.4%	18 $\rightarrow$ 23
SmolLM2-360M	none	5.6%	27.8%	32.2%	144 $\rightarrow$ 225
SmolLM2-360M	L1+L2	5.6%	11.1%	6.7%	8 $\rightarrow$ 9
Qwen2.5-1.5B	none	8.9%	23.3%	30.0%	106 $\rightarrow$ 162
Qwen2.5-1.5B	L1+L2	3.3%	10.0%	15.6%	24 $\rightarrow$ 32

Observed over 90 simulated days, so each share has a standard error of about 2-5 percentage points. L1+L2 is the best cascade by the objective of Section 4.4 within the 500 ms budget; it is not the minimum-dollar cascade of Section 5.5.

The gap shrinks with guardrails but does not vanish: for the best cascade, the Poisson plan is exceeded on 7-16% of days at $c = 0.44$, against 30-32% without guardrails. This follows Proposition 2 (for given c , $\phi = 1 + \lambda c^2$ is smaller when the event rate λ is smaller, and the Fano factor of the request stream is diluted by thinning, Proposition 3): over-dispersion inflates the safety stock less in absolute terms (about 5, 1, 8 extra units) although the exceedance probability stays between 1.3 and 3 times the target.

5.5 Capacity, latency and cost by cascade

Table 7 gives, for each generator, the residual rate λ , the capacity k^* , the added latency and the false-block volume of the main cascades at 200 queries/day, with bootstrap intervals over the evaluated pool. Intervals are wide at this pool size (300 questions), so differences of a few units of k^* between cascades are not significant; the large differences are.

Table 7. Cascade performance at 200 queries/day, $\alpha = 0.95$, SLA 500 ms mean added latency (95% bootstrap CI in brackets)

Generator	Cascade	λ per day	k^*	Mean added latency (ms)	False blocks / day	System failures / day
Qwen2.5-0.5B	none	120.7 [109, 132]	139 [126, 151]	0	0.0	0.0
Qwen2.5-0.5B	L1+L2	12.7 [7, 18]	19 [12, 25]	97	12.7	0.0
Qwen2.5-0.5B	L2	12.7 [7, 18]	19 [12, 25]	98	12.0	0.0
Qwen2.5-0.5B	L3	50.7 [39, 60]	63 [49, 73]	1759 $\times$	19.3	60.0
Qwen2.5-0.5B	L2+L3	11.3 [7, 17]	17 [11, 24]	287	16.7	6.0
SmolLM2-360M	none	130.0 [120, 142]	149 [138, 162]	0	0.0	0.0
SmolLM2-360M	L1+L2	4.0 [1, 7]	8 [3, 12]	93	48.7	0.0
SmolLM2-360M	L2	4.0 [1, 7]	8 [3, 12]	97	48.7	0.0
SmolLM2-360M	L3	67.3 [58, 78]	81 [71, 94]	1780 $\times$	11.3	50.0
SmolLM2-360M	L2+L3	4.0 [1, 7]	8 [3, 12]	249	50.7	6.7
Qwen2.5-1.5B	none	94.0 [82, 106]	110 [97, 123]	0	0.0	0.0
Qwen2.5-1.5B	L1+L2	17.3 [12, 24]	24 [18, 32]	87	14.0	0.0
Qwen2.5-1.5B	L2	17.3 [12, 24]	24 [18, 32]	88	14.0	0.0
Qwen2.5-1.5B	L3	38.7 [29, 48]	49 [39, 60]	1596 $\times$	20.7	49.3
Qwen2.5-1.5B	L2+L3	16.7 [11, 23]	24 [16, 31]	279	19.3	6.7

L2+L3 is the routed cascade (the judge L3 is called only if the L2 score is below 0.9); L3 alone runs the judge on every answer. $\times$ = violates the 500 ms budget. Mean added latency is per query, including abstentions (which bypass the guardrails), so it differs from the per-layer means of Table 5; each judge call is capped at the 2 s timeout, after which its verdict is discarded and it counts as a system failure. L1+L2 and L1+L2+L3 equal L2 and L2+L3 up to small differences because L1 is nearly inert on this task.

Capacity. A single task-specific verifier (L2) reduces required capacity from 139, 149, 110 to 19, 8, 24 units/day (a 78-95% reduction). Adding the routed judge changes it to 17, 8, 24, i.e. by 2, 0, 0 units, while multiplying added latency by about three and adding roughly 6 system failures per day through judge timeouts. Running the judge on every answer without routing is infeasible under any latency budget we consider (about 1.6-1.8 s per query) and produces about 50-60 system failures per day on our CPU setup.

Latency budget. Under a 100 ms budget only L1 and L2 (L2 takes about 90-100 ms here) are feasible and no judge layer fits; under 500 ms the routed judge fits on average but not at the 95th percentile (1.8-1.9 s). Latencies are CPU-bound and would be much lower on GPUs or hosted inference.

Selection and cost. The best cascade by the objective of Section 4.4 is L1+L2 for all three generators (a latency-penalised objective with arbitrary weights; the minimum-dollar cascade below was not replayed under bursty traffic). At the scenario costs (C_h = \$150 per capacity unit, \$1 per falsely blocked answer, \$100 per hallucination reaching a user; illustrative values) the minimum-dollar cascade is L1+L2+L3 for Qwen2.5-0.5B, L1+L2 for SmolLM2-360M, L1+L2+L3 for Qwen2.5-1.5B. The daily cost falls from \$32,917, \$35,350, \$25,900 without guardrails to \$3,700, \$1,649, \$5,286 at the minimum. The paired pool-

bootstrap interval of the daily saving from adding the routed judge to L1+L2 is \$429 [-\$5, \$931]; -\$2 [-\$5, \$0]; \$61 [-\$9, \$646] for the three generators: positive but not distinguishable from zero for the two Qwen generators, and slightly negative for SmolLM2-360M, whose classifier leaves only 6 of its 195 hallucinations for the judge to catch (none were caught). Under this fail-open policy the judge therefore does not demonstrably pay for itself for any generator. **Sensitivity to the timeout policy:** if slow judge verdicts were waited for and credited (no 2 s timeout), the stand-alone clearance of L3 would be 63-84% (against 48-59%), the routed cascade would need $k^* = 15, 8, 21$, and the saving of adding the judge would be \$929 [\$274, \$1,861]; -\$2 [-\$5, \$0]; \$709 [\$130, \$1,575], so the judge's apparent value depends on accepting latencies above 2 s. The residual hallucinations after L2 are few (19, 6, 26 of 300 questions), the 27-scenario grid of Section 4.4 varies cost parameters and does not reflect sampling error, and the three runs share their questions, so none of these differences between generators is a general finding. The classifier is the dominant layer, the deterministic layer is nearly free but also nearly useless on this task, and the judge's marginal value is small and uncertain.

5.6 Evidence from public production traces

Request arrivals (BurstGPT). The Azure OpenAI trace has 10.6 million requests over 231 recorded days; 4.6% of requests failed (zero response tokens). The hour-of-day profile has a peak-to-trough ratio of 2.2 (peak in the late afternoon and evening hours, trough in the morning hours; the clock offset of the trace is unknown), so the arrival process is non-homogeneous, as the NHPP formulation assumes. Within an hour, counts are more variable than Poisson: the variance-to-mean ratio of counts per bin (Poisson = 1; median over hours with a mean of at least one request per bin, 919, 2477, 3984 and 4999 hours at the four scales) is 1.4 at 1 s, 5.3 at 10 s, 3.9 at 1 min and 5.1 at 10 min, i.e. requests are clustered at every scale from 1 s upward (mildly at 1 s, strongly from 10 s). By Proposition 3 an unprotected hallucination stream with $p = 0.6$ would inherit about $1 + 0.6(F - 1) \approx 3.5$ of that dispersion at the 10-minute scale, whereas a cascade with residual probability $p = 0.05$ would retain about 1.2.

Day-to-day volume is extremely variable: the median day has 19,024 requests, with a range from 52 to 425,604. The median absolute day-to-day log change is 0.61 (a factor of 1.8), 73% of days differ from the previous day by more than 30%, and the robust log-standard deviation around a 15-day rolling median is 0.80. Net of the local level (day count over the mean of the 14 neighbouring days) the coefficient of variation is 1.22 in days 0-120 and 2.77 in days 225-334. The smallest coefficient of variation of any of the 163 14-day windows with all days above 1,000 requests is 0.44 (mean 140k requests/day; the median window has 0.91), so 0.44 is a **floor** for the variability of this trace and not an estimate comparable to the net-of-trend values of Section 5.7. The variability reflects traffic growth, campaigns, regime shifts and possibly logging gaps and not only random fluctuation (the series also contains 7 near-duplicate day pairs, which we treat as a data artefact), so we use it as a stress test, not as an estimate of the dispersion of hallucination counts. As a stress test we replay the measured multipliers (rescaled to mean 1 so that λ is the true mean rate) on an event stream with mean $\lambda = 10$ and 120 events/day: the Poisson plan (15 and 138 units) is exceeded on 23% and 31% of days with the multipliers of days 0-120 (CV 1.2) and on 16% and 23% with those of days 225-334 (CV 2.8, inflated by regime shifts), against a target of 5%, and the empirical 95% quantiles of the replayed counts are 30, 332, 29, 334 units ($\lambda = 10$ and 120 in each period) instead of 15, 138, 15, 138.

Service incidents (LLM providers). Between March 2023 and August 2024 the three providers declared 457 incidents (0.83 per day; OpenAI 0.51, Anthropic 0.26). Incident starts are strongly non-homogeneous: the mean weekday count is 4.1 times the mean weekend count and hour-of-day homogeneity is rejected ($p < 10^{-4}$ for every provider and subset). A homogeneous Poisson model is rejected on every inter-arrival and dispersion test. Allowing for weekly and daily seasonality removes part of the lack of fit but not all of it: the daily index of dispersion is 1.53 ($p = 3 \times 10^{-14}$) and still 1.30 after stratifying by day of week ($p = 6 \times 10^{-6}$); for OpenAI alone the time-rescaled inter-arrival test is consistent with an NHPP ($p = 0.23$) but its stratified index is still 1.19 ($p = 0.003$), while for Anthropic and for the major and critical subset both are rejected. A negative-binomial fit to daily counts gives $r \approx 1.6$. Part of this over-dispersion may come from a changing incident rate over the 18 months, which a weekday-stratified test does not remove (Section 5.8 adjusts the baseline for it). The practical consequence at this low rate is small: the Poisson and negative-binomial 95% capacities coincide (3 incidents per day) and the Poisson capacity is exceeded on 3.5% of days, below the 5% target. This agrees with Proposition 2: over-dispersion is statistically clear but $\phi = 1 + \lambda c^2$ remains modest (about 1.2-1.5 here, from the stratified indices) when λ is below one per day, and it becomes consequential only at higher rates.

Human response proxy. Incident duration (declared start to resolution) has a median of 1.0 h, a 90th percentile of 5.1 h and a 99th percentile of 34 h; the mean is 3.1 h and the coefficient of variation 3.5, i.e. response times are heavy-tailed, not exponential (12.3% are zero-duration notices). The time-averaged number of concurrently open incidents is 0.11, which equals the M/G/ ∞ mean by Little's law and therefore is not a test; the more informative comparison is that the 95th percentile of the open count is 1, as for a Poisson occupancy with that mean (1), and that two or more incidents are open simultaneously 0.7% of the time. At provider level a single standing responder therefore covers 95% of the time, whereas the heavy tail of durations explains the rare long overruns. These are provider-declared outage incidents (elevated errors, latency, degraded service), not hallucinations, so they inform the arrival and service-time structure of an LLM incident stream but not its rate.

Dated hallucination cases (court decisions). The only public, dated log of hallucination incidents we found is a curated database of court decisions in which a party relied on AI-fabricated content (Charlotin; 2,149 dated cases, 2023-04-14 to 2026-10-01). It is legal-domain only, the date is the decision date at day resolution, and detection and reporting bias is strong: counts grew from a handful per month to over a hundred in 2025, so only the flatter period from October 2025 to August 2026 (1,527 cases, 6.3 per weekday, weekend rate 0.04 of the weekday rate) is analysed; September 2026 (95 cases) and October 2026 (1 case) are excluded because recent months are incomplete (reporting lag). In that period the weekday counts are over-dispersed relative to Poisson (day-of-week-stratified index 1.40, $p = 1 \times 10^{-4}$; daily-rate CV net of trend 0.24), but part of this comes from the 10 US federal-holiday weekdays (mean count about 1.6 against 6.3); without them the index is 1.25 ($p = 0.012$) and the CV 0.18, a weak over-dispersion. At this rate the effect on capacity is small: the 95% capacity is 11 under Poisson and 12 under a negative binomial with the stratified dispersion index; a Poisson plan against the local mean is exceeded on 6.3% of weekdays (5.6% with a causal trailing-mean plan; 4.4% and 5.9% without the holidays). This is consistent with Proposition 2 and with the other low-rate streams, and it is, to our knowledge, the only dispersion estimate for dated hallucination incidents that public data allow, with the caveats above.

5.7 Human triage workloads in public company ticket data

The fourth family of public data concerns what happens after an incident is detected: how tickets arrive at, and are worked by, human teams in real organisations. The streams are customer-support and IT-incident tickets, not LLM incidents, but they are the best available public evidence on the arrival variability and the service-time distribution of human triage. The ServiceNow log is useful for arrivals only in its middle weeks (98.9% of its incidents are opened between 29 February and 1 June 2016), so that stream is a 12-week segment. Holidays and bridge days appear as isolated low-count days that a local mean cannot absorb; each stream is therefore analysed twice, with all work days and with such days excluded by the rules of Section 4.5 (arrow notation: all days $\rightarrow$ excluded).

Table 8. Arrival structure of human-handled ticket streams (work days only; Poisson plan = 95% quantile of Poisson with the local mean)

Stream	Tickets (per work day)	Work-day-stratified dispersion index (includes slow trends)	Daily rate CV net of trend and weekday	Poisson plan exceeded: in-sample local mean	NB plan exceeded: in-sample	Poisson plan exceeded: causal	NB plan exceeded : causal
Software company (customer tickets 2017-23)	22,016 (13.1)	3.5 $\rightarrow$ 3.1	0.32 $\rightarrow$ 0.27	9.0% $\rightarrow$ 7.8%	3.4% $\rightarrow$ 4.0%	9.6% $\rightarrow$ 8.6%	1.9% $\rightarrow$ 2.5%
ServiceNow incidents (12-week segment, 2016)	22,035 (350.2)	25.2 $\rightarrow$ 5.9	0.25 $\rightarrow$ 0.10	35.0% $\rightarrow$ 20.0%	1.7% $\rightarrow$ 5.5%	15.0% $\rightarrow$ 6.7% (20 $\rightarrow$ 15 days)	0.0% $\rightarrow$ 0.0%
Italian software help desk (2010-12)	3,804 (5.3)	2.5 $\rightarrow$ 2.4	0.41 $\rightarrow$ 0.38	7.5% $\rightarrow$ 7.2%	3.7% $\rightarrow$ 3.8%	9.7% $\rightarrow$ 8.7%	2.9% $\rightarrow$ 2.8%

Each cell: all work days $\rightarrow$ low-volume or holiday days excluded (software company and ServiceNow: days below 35% of the rolling median, 62 and 5 days; Italian help desk: national holidays, 26 days). Target exceedance is 5%. The dispersion index includes slow trends, whereas the CV is net of a local level, so the two are not directly comparable. In-sample: the local mean is a two-sided leave-one-out mean over about 40 surrounding work days (eight weeks) after removing the weekday effect, and c is estimated from the same days that are scored, so neither is a feasible forecast. Causal: trailing 20-work-day mean, c estimated on earlier days only, rolling origin (the weekday effects are estimated on the full sample). The software company works Sunday to Thursday (weekend = Friday and Saturday); the other work weeks are Monday to Friday. A High/Highest/Blocker subset of the software company is not analysed because its rate rises from about zero to eleven tickets per work day over 2017-2023 (a change in the priority field).

Arrivals (Table 8). All streams are strongly non-homogeneous over the week and the day (hour-of-week homogeneity rejected, $p < 10^{-100}$; weekend ticket rates are 0.09, 0.12 and 0.04 of work-day rates), and arrivals are far from Poisson even after time-rescaling ($p < 10^{-90}$ for the two long streams), because daily volumes vary beyond the weekly cycle. Net of slow trends and the weekday effect, the coefficient of variation of the daily rate is 0.32 $\rightarrow$ 0.27 (software company), 0.41 $\rightarrow$ 0.38 (Italian help desk) and 0.25 $\rightarrow$ 0.10 (ServiceNow segment), where holidays and bridge days account for most of the ServiceNow value; the dated hallucination cases of Section 5.6 give 0.24 $\rightarrow$ 0.18. These values are of the same order as the scenario values of Section 5.4 ($c = 0.3$ lies within the range of 0.10-0.38 after exclusions; $c = 0.44$ lies above it), although all of them come from non-LLM or legal-domain streams and we do not claim a universal value. Because $\phi = 1 + \lambda c^2$, the effect on capacity depends on the rate: against the in-sample local mean, a Poisson-sized plan is exceeded on 7.8%, 7.2% and 20.0% of work days (about 14, 5 and 376 tickets per day) instead

of 5% (9.0%, 7.5% and 35.0% with all days). A causal plan (trailing mean and previous-days c , which is what an operator could use) is exceeded on 8.6% and 8.7% of days for the two long streams with the Poisson quantile and 2.5% and 2.8% with a negative-binomial quantile, i.e. the negative-binomial plan is somewhat conservative for reasons we cannot identify from these data. For the 12-week ServiceNow segment the causal evaluation covers only 15 days and is not informative. The negative-binomial results in the in-sample columns are a calibration check of Proposition 2 rather than an out-of-sample test, since c is estimated from the same days that are scored; the causal columns are the feasible comparison.

Table 9. Resolution and handling times of tickets (hours)

Stream / measure	n	Median	Mean	90th pct	99th pct	CV	Best fit (AIC) / Weibull shape
Software company: creation to resolution	22,016	167.8	698.8	1487	8806	3.0	weibull / 0.51
Software company: time in "in progress"	19,662	2.2	97.3	220	1532	3.6	weibull / 0.30
ServiceNow: opened to resolved	23,323	22.2	178.5	382	2849	3.0	weibull / 0.36
ServiceNow: Critical/High only	496	49.7	172.9	391	1086	2.0	weibull / 0.55
Italian help desk: first to last event	3,803	94.4	211.2	627	1008	1.3	weibull / 0.50

Calendar hours including waiting states (customer, vendor, queue); the in-progress measure counts only time in that workflow state. Exponential fits are rejected in every case ($KS\ p < 10^{-39}$).

Service times (Table 9). Resolution times are heavy-tailed and far from exponential: median 168 h against a mean of 699 h for the software company, 22 h against 178 h for ServiceNow and 94 h against 211 h for the Italian help desk, with coefficients of variation of 3.0, 3.0 and 1.3 (an exponential has 1) and Weibull shapes of 0.36-0.51, i.e. a decreasing hazard: a ticket that has stayed open for long is likely to stay open much longer. The same holds for the time a ticket spends in the "in progress" state (median 2.2 h, mean 97 h, CV 3.6). The Critical/High incidents of ServiceNow take longer (median 50 h) and are also heavy-tailed. Two consequences for capacity planning follow. First, models that assume exponential service (M/M/c) are not supported, which is a further reason to size capacity from the quantile of the arrival-count distribution and to treat the service-time tail explicitly rather than through a mean-based utilisation. Second, in the one stream with handling-time data (the software company; workload = the in-progress hours of the tickets created that day, attributed to the creation day, Sunday to Thursday only) the 95th percentile of the daily workload is 3.2 times its mean, against 1.9 for the daily ticket count (CVs 1.27 and 0.47); capacity measured in handling hours thus needs a larger multiple of the mean than capacity measured in arrivals, and the "unit of triage capacity" of Section 3.2 must be defined for a typical ticket with the workload tail added separately. The in-progress time is calendar time in a workflow state, not effort, and the comparison is not detrended, so the ratio is indicative only.

Concurrency (descriptive). For the software company and the Italian help desk the time-average number of open tickets is 290 and 34, with variance-to-mean ratios of 21 and 2.6 (Poisson occupancy has 1). The observed 95th percentile of open tickets is 384 and 51, against 319 and 44 for a Poisson (M/G/ ∞) occupancy with the same mean, i.e. 20% and 16% higher. This is a statement about a static backlog plan sized with the long-run mean, not a test of Poisson arrivals or of the service-time structure: both logs change in volume over their span (and in service time for the Italian desk), and a slowly varying volume alone produces such an excess, so the comparison is descriptive. The first and last 5% of time were removed to limit edge effects, and the ServiceNow log is not used here because it starts and stops abruptly.

5.8 Typed failure models on public production data

We now fit the process classes of Section 3.6 to the two public failure datasets (the numbers in this section are produced by `run_failure_models.py`).

Table 10. Seasonal NHPP versus seasonal Hawkes process for provider-declared incidents (March 2023 - August 2024)

Stream	Incidents	Baseline	Branching ratio α	Mean cluster size $1/(1-\alpha)$	Excitation half-life (h)	AIC gain over NHPP	LR test vs NHPP (bootstrap p)	Rescaled-gap KS p: NHPP $\rightarrow$ Hawkes
all providers	457	profile, constant level	0.16	1.19	7.1	19.7	23.7 (≤ 0.005)	$5 \times 10^{-4} \rightarrow 0.47$
		profile $\times$ monthly level	0.07	1.07	2.9	3.5	7.5 (0.025)	0.166 $\rightarrow$ 0.91
OpenAI	280	profile, constant level	0.11	1.12	3.8	10.8	14.8 (≤ 0.005)	0.076 $\rightarrow$ 0.26
		profile $\times$ monthly level	0.05	1.06	1.5	2.6	6.6 (0.020)	0.135 $\rightarrow$ 0.82
Anthropic	141	profile, constant level	0.21	1.27	4.2	26.2	30.2 (≤ 0.005)	$1 \times 10^{-6} \rightarrow 0.13$
		profile $\times$ monthly level	0.09	1.10	3.8	2.2	6.2 (0.025)	0.016 $\rightarrow$ 0.19
major/critical (impact ≥ 2)	198	profile, constant level	0.13	1.14	5.8	9.4	13.4 (≤ 0.005)	$5 \times 10^{-4} \rightarrow 0.13$
		profile $\times$ monthly level	0.06	1.06	1.5	3.9	7.9 (0.015)	0.115 $\rightarrow$ 0.44

Baseline: weekday/weekend $\times$ hour-of-day profile estimated from the incident data and held fixed; second row of each stream: additionally multiplied by a monthly level estimated from the monthly counts. Exponential excitation kernel; bootstrap p from 200 parametric simulations under the corresponding NHPP (smallest possible value 0.005). KS: Kolmogorov-Smirnov test of time-rescaled gaps against $\text{Exp}(1)$ with the fitted compensator; these p-values are optimistic because the profile (48 slots), the monthly levels and the Hawkes parameters are all fitted on the same events.

Provider incidents: weekly seasonality, a drifting level and mild self-excitation. The incident rate was not constant over the 18 months: monthly counts were 11-27 in the first twelve months and 27-54 afterwards (homogeneity across months rejected, $\chi^2 = 87$ on 17 df). With a fixed seasonal profile and a constant level, a self-exciting term improves the fit for every stream (AIC gains of 9-26; the bootstrap rejects the NHPP for all four streams; branching ratio 0.11-0.21), but part of this is the slowly changing level, which the self-excitation term can mimic. With the baseline multiplied by a monthly level, the estimated branching ratio falls to 0.05-0.09 (mean cluster 1.06-1.10 incidents, half-life 1.5-3.8 h), the AIC gains shrink to 2-4, the bootstrap p-values are 0.015-0.025, and the (optimistic) time-rescaling test no longer rejects the NHPP for the combined stream ($p = 0.17$), OpenAI and the major and critical subset, so a weekday/hour profile with a drifting level plausibly describes most of the data. A weak residual self-excitation remains (roughly one incident in 11 to 18 followed by a clustered successor, with borderline evidence), and Anthropic's time-rescaling test still rejects the NHPP ($p = 0.016$). The implied long-window Fano factor of 1.15 (Proposition 6b) is below the stratified dispersion index of 1.30 of Section 5.6, so part of the measured over-dispersion is the drifting level. The conclusion for planning is modest: provider incidents need a seasonal, level-adjusted baseline; whether they also cluster is weakly supported and the effect is small at this rate.

Serving failures: heavy-tailed episodes. In the BurstGPT trace, failed requests are concentrated: 3.6% of active minutes (outage-like episodes, defined as minutes with more than 20% failed requests and at least five failures, merged when ten minutes apart) contain 68.1% of all 487,629 failures (26-96% across the five definitions of Table 11). The 672 episodes have a median duration of 4 minutes (90th percentile 27, maximum 455) and heavy-tailed sizes: median 78 failed requests, mean 494, 99th percentile 10,419, maximum 27,908 (CV 4.7). A lognormal fit has $\sigma = 1.70$; the Hill estimator on the upper 20% gives a tail index of about 0.9, which at $n = 672$ cannot discriminate a Pareto tail with infinite variance from the lognormal, so we do not read it as evidence of an infinite variance. What matters for Proposition 6c is the second-moment ratio $E[X^2]/E[X]$ of the episode sizes, which is 11,374 empirically and 5,838 under the lognormal fit: the Fano factor of failed-request counts that arrive in such episodes is in the thousands. Because request volume itself ranges from 52 to 425,604 per day, we also condition on volume: the Pearson dispersion of daily failed-request counts given the daily request volume (expected count proportional to volume, 231 days) is 5,961, far from the value of about 1 of a Poisson-thinned stream, so a Poisson description of failed requests is wrong by several orders of magnitude even after allowing for traffic. The episodes themselves are not Poisson either: they average 2.9 per day; the daily dispersion index is 19.9 unconditionally and 16.3 given daily volume (123 days with an expected count of at least one); inter-arrival times have a coefficient of variation of 4.2; and a Hawkes process fits much better than an NHPP when the baseline is the hour-of-day profile (AIC 4143 $\rightarrow$ 2374; branching ratio 0.85, half-life 2.4 h) and also when the baseline is proportional to the hourly request volume (AIC 3366 $\rightarrow$ 2326; branching ratio 0.74), although the rescaled-gap test still rejects the Hawkes fit ($p = 4 \times 10^{-7}$ and 9×10^{-10}). The branching ratio depends on how episodes are delimited (Table 11): it stays between 0.52 and 0.85 for five definitions and the heavy tail of sizes persists, but the unconditional daily dispersion index of episode counts ranges from 2.1 to 19.9, and a long outage may be split into several fragments, so only the direction (clustering and heavy tails) is robust.

Table 11. Sensitivity of the BurstGPT failure-episode model to the episode definition

Failure-share threshold	Min. failures per minute	Merge gap (min)	Episodes	Share of failures	Daily dispersion index	Hawkes α	Median / 99th pct size
20.0%	5	10	672	68.1%	19.9	0.85	78 / 10419
20.0%	5	60	211	88.9%	2.1	0.74	80 / 32372
10.0%	5	30	279	96.5%	3.2	0.66	61 / 30295
50.0%	5	10	192	25.6%	3.3	0.52	43 / 19484
20.0%	20	30	154	74.4%	5.2	0.78	259 / 32333

The first row (20%, 5, 10) is the definition used in the text. Hawkes fits use 30 bootstrap replications and hour-of-day baselines.

Capacity under the typed model. To see what the typed description changes for a human-triage capacity plan, we combine three human-handled classes for an illustrative organisation with parameters taken from our estimates and independent classes: hallucinations after the classifier-only cascade L1+L2 ($\lambda_H = 12.7$ per day, Qwen2.5-0.5B run; the daily-rate CV c_H is assumed in the range 0.3-0.45, near the 0.27 and 0.38 (0.32 and 0.41 with all days) estimated net of trend for the two long ticket streams, so $\phi_H = 1 + \lambda_H c_H^2 = 2.14$ -3.57 by Proposition 2), provider incidents affecting the organisation ($\lambda_P = 0.83$, $\phi_P = 1.30$ from the day-of-week-stratified index) and own serving-failure episodes (λ_S from the episode counts over the 231 recorded days; $\phi_S = 2.1$ for the coarse definition, 19.9 for the fine one, and 1.8 for the coarse definition

conditioned on request volume). Treating the total as one Poisson stream gives a 95% capacity of 23 (fine definition) or 21 (coarse) units per day at $c_H = 0.37$; the typed superposition requires 33 and 25, and the single-Poisson plan is exceeded on 15.4% and 12.6% of days instead of 5% (Monte Carlo of 400,000 days; effective dispersion $\phi_{\text{eff}} = 2.6\text{-}5.7$, Proposition 6a). Across the nine scenarios (c_H of 0.3, 0.37 and $0.45 \times$ three treatments of the serving class) the required capacity is 24-34 units against a single-Poisson plan of 21-23, i.e. 14-48% more, with the single-Poisson plan exceeded on 10.4%-17.2% of days. The hallucination class alone, whose dispersion is assumed, accounts for a gap of 21% (23 rather than 19 units); the provider class adds little, and the serving class widens the gap substantially only when it is treated as strongly over-dispersed (the fine episode definition, $\phi_S = 19.9$), while with the coarse definition, also when conditioned on request volume, the total gap is about that of the hallucination class. The scenario is therefore an illustration of the size of the single-Poisson error under stated assumptions, not evidence for the typed model: class means are not those of any real organisation, the comparator is a single Poisson (a negative-binomial plan on the total would close part of the gap), and a serving outage would plausibly raise the hallucination and ticket volumes at the same time, which would widen it.

6. Discussion

6.1 Implications for IS practice and research

- **Size capacity from a quantile, and fit the dispersion first.** Provisioning at the mean load (k = round of the estimated rate) failed on 28-49% of days across the configurations we replayed. The Poisson quantile fixes this when daily counts are Poisson, but Proposition 2 and the burst replays show that a modest day-to-day variation in demand (CV 0.3-0.44) raises the exceedance rate of a Poisson-sized plan from 5% to 23-28% ($c = 0.3$) and 30-32% ($c = 0.44$) without guardrails; this follows from the injected multiplier and confirms the proposition on measured outcomes. Real LLM request streams vary far more than that (a stress test, Section 5.6). Dispersion should be estimated from incident logs before the plan is fixed. Over-dispersion matters most for large incident streams (weak guardrails) and least for small ones (Propositions 2 and 3): provider outages at about 0.8 per day and dated hallucination cases at about 6 per weekday were statistically over-dispersed but changed the 95% capacity by at most one unit, whereas the ServiceNow segment (about 350 tickets per work day; Poisson plan exceeded on 20% of work days even after excluding holidays and bridge days) and the unprotected hallucination stream of 100+ events per day were affected strongly.
- **Guardrail value is an empirical, per-deployment quantity.** The layers' stand-alone clearance, their dependence, their false-block cost and their latency vary strongly by layer and model. In our data one task-specific verifier delivered most of the benefit (a 78-95% reduction in required capacity at the price of 15-71% of correct answers blocked), the deterministic layer was nearly inert, and the routed judge's gain was small (k^* reduced by 2, 0, 0 units) and not distinguishable from zero in cost terms (paired intervals for the daily saving \$429 [-\$5, \$931]; -\$2 [-\$5, \$0]; \$61 [-\$9, \$646]), at about three times the latency and under a fail-open 2 s timeout. A fixed "three layers" prescription is not supported by these data; the method here selects the stack from measurements. A classifier that is not validated can be actively harmful: our NLI layer had chance-level discrimination and appeared to cut capacity only because it blocked almost everything.
- **Model the failure classes separately and combine them at the ticket level.** Treating all failures as one Poisson stream misdescribes each class in a different way: hallucination counts are over-

dispersed through demand variability, provider incidents are seasonal with a drifting level and at most a weak clustering, serving failures arrive in heavy-tailed episodes (68.1% of failed requests fall in 3.6% of the active minutes), and human triage has heavy-tailed service. In our illustrative organisation a typed superposition required 14-48% more daily capacity than a single Poisson on the total across nine scenarios; the gap is driven by the assumed over-dispersion of the hallucination class and by how strongly over-dispersed the serving class is taken to be, so the example shows the size of the single-Poisson error under stated assumptions rather than a validated gain.

- **Human triage is bursty and heavy-tailed in real organisations.** In public ticket streams the day-to-day coefficient of variation of the arrival rate was 0.10-0.38 net of trend after excluding holiday and low-volume days (0.25-0.41 with all days), a Poisson-sized plan against the in-sample local mean was exceeded on 7-20% of work days instead of 5% (the two long streams: 8.6% and 8.7% with a causal plan), resolution times had coefficients of variation of 1.3-3.0 with decreasing hazards, and in the one stream with handling-time data the 95th percentile of the daily workload was 3.2 times its mean against 1.9 for the arrival count. Capacity in *arrivals* is therefore not capacity in *hours*, and the tail of the service time needs its own headroom.
- **Routing is a first-order design choice.** Sending only low-agreement answers to the expensive judge cuts judge calls by roughly an order of magnitude, but it also forgoes catches among answers the classifier rated as consistent; the share of hallucinations passing the L2+L3 cascade was 1.9-2.3 times the independence prediction with routing against 1.3-1.6 times without it (Section 5.2).
- **Governance.** Because k^* converts a detector's precision and recall into headcount, the framework gives IT governance a common unit (units of triage capacity or dollars per day) in which to compare detectors, set the SLA and justify standing on-call capacity.

6.2 Limitations

- **Simulated arrivals and no operator log of hallucinations.** Outcomes are measured on real model outputs, but arrival times are simulated and each arrival replays a bootstrapped example, so the Poisson and time-rescaling checks of Section 5.3 verify the machinery and are partly by construction. The public traces we use (BurstGPT, the provider incident history, the ticket logs) are real production data but record request arrivals, service incidents and human tickets, not hallucinations. The only dated hallucination log we found is a curated database of court decisions (Section 5.6), which is domain-specific, at day resolution, and subject to strong detection and reporting bias. To our knowledge no public production log of hallucination incidents from an operator exists. The burstiness of the replay is therefore a scenario informed by, not estimated from, real streams. Validation on proprietary incident logs is the main follow-up.
- **The detection assumption.** The model assumes that every hallucination that passes the guardrails is detected and becomes exactly one human-handled event. In practice only a fraction is noticed or reported, and detection is a failure point of its own; the capacities we report are upper bounds that hold for full detection (Section 3.1).
- **Small sample, shared questions, post-hoc layer change.** The three generators answered the same 300 questions, so their results are paired and not independent replications, and each layer estimate rests on 141-195 hallucinations and a few to a few dozen residual ones. The extractive-QA verifier was introduced after the NLI classifier proved uninformative, so the L2 choice was made after seeing results on the same data; it is trained on SQuAD, which flatters it relative to a domain-

agnostic classifier, and its threshold was not tuned. The judge is Qwen2.5-1.5B in every run, so one arm is self-judged. The analyses should be repeated on a held-out sample with a pre-registered layer configuration.

- **Dependence estimates are imprecise, and the timeout policy matters.** The positive association between L2 and L3 and the excess over the independence formula are weak, their bootstrap intervals include zero (or one) for two of the three generators, and the independence prediction is computed from the same sample's marginals. The judge results assume that verdicts of calls slower than 2 s are discarded (fail-open); crediting them would raise the judge's clearance and its apparent value (Section 5.5).
- **Typed models are illustrative in combination.** The Hawkes and episode fits use one kernel family (exponential), a fixed seasonal profile estimated from the same data, and, for BurstGPT, an episode definition that matters quantitatively (Table 11) and may split a long outage into fragments; the bootstrap p-values are bounded below by $1/(B + 1)$ and the time-rescaling p-values are optimistic. The capacity comparison of Section 5.8 combines classes from different organisations and datasets under independence, with an assumed hallucination dispersion and a single Poisson as the comparator; common-cause dependence between classes would widen the gap.
- **Ticket data are not LLM incidents.** The ticket streams are customer-support and IT incidents; whether LLM-hallucination tickets share their arrival variability and service times is an assumption (plausible for the daily-rate CV, which is invariant to independent thinning with constant p , less so for service times, which depend on the verification task). Resolution times are calendar times that include waiting for customers or vendors and are not triage effort; the in-progress state time is a closer but still imperfect proxy. Ticket volumes in two of the logs change over their span, which we address by estimating variability against a local mean, the ServiceNow log can be used only for a 12-week segment, and the arrival-variability estimates of Section 5.7 rest on three streams.
- **Isolated low-count days.** Holidays, bridge days and outages produce single-day dips that an eight-week local mean cannot absorb and that inflate the dispersion and the exceedances of the in-sample plans (Section 5.7); we report results with and without such days, using a volume rule for two streams and holiday calendars for two others, and the rule is outcome-based and therefore not applied to low-rate streams. The unexplained conservatism of the causal negative-binomial plan shows that our variability estimates are not fully understood.
- **Public-trace artefacts.** BurstGPT has no calendar dates, a 104-day gap, near-duplicate day pairs and order-of-magnitude regime changes in daily volume, so its day-level variability reflects growth and logging as well as randomness; we use it as a stress test and its smallest 14-day coefficient of variation as a floor, not as an estimate. The provider incident data are provider-declared, with minute resolution and detection and reporting bias; incident duration is the declared start-to-resolution time, not triage effort.
- **Small open models and one task.** We study 0.4-1.5B-parameter models on extractive QA with unanswerable questions. Absolute hallucination rates are much higher than for frontier models, and the layers are CPU-bound; latencies on GPU or hosted inference differ. The *method* transfers, the *numbers* do not.
- **Label noise and cost parameters.** Ground truth uses string and token-F1 matching against SQuAD gold answers, which can mark valid paraphrases as hallucinations and inflates the base rate. The

cost parameters and selection weights are assumptions; only the labour-to-compute ratio and the sensitivity grid are informative, and the grid does not reflect sampling error.

7. Conclusion

Treating human-handled LLM incidents as a counting process turns guardrail performance into a staffing and cost decision. Under the stated assumptions, capacity should be sized from an over-dispersion-aware quantile of the daily count; the multiplicative guardrail formula is an optimistic bound whose gap must be measured; a single Poisson on all failures misdescribes every class (hallucinations are over-dispersed through demand variability, provider incidents are seasonal with a drifting level and weak clustering, serving failures are heavy-tailed episodes); and the cost-minimising number of layers depends on integer capacity effects, the floor on standing staff, latency, the labour-to-compute ratio and the measured clearance, and is not fixed at three. On measured outputs of three small open models answering the same 300 questions, a single task-specific verifier reduced the required capacity by 78-95% relative to no guardrails while blocking 15-71% of correct answers; a routed judge did not demonstrably pay for itself under a 2 s fail-open timeout and added system failures and about three times the latency, and every cascade containing the verifier blocked correct answers. The share of hallucinations passing a two-layer cascade was 1.3-1.6 times the independence prediction without routing and 1.9-2.3 times with routing, with wide intervals. Mild burstiness (daily-rate CV of 0.3-0.44, a range informed by public ticket streams and a production trace) raised the exceedance rate of a Poisson-sized plan from 5% to roughly 23-32% without guardrails and 7-16% with the best cascade. Public production data corroborate the structure: an LLM request trace shows clustered arrivals from the 10-second scale upward, provider incident streams are strongly seasonal with a drifting level and over-dispersion that remains after weekday stratification, a log of dated hallucination cases shows weak over-dispersion at about six cases per weekday, and company ticket streams show a net-of-trend daily rate CV of 0.10-0.38 (0.25-0.41 with all days) and heavy-tailed resolution times. Hallucination arrivals themselves are simulated and no operator log of hallucinations is public, so validating the counting-process assumptions on production incident logs is the key next step.

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